COMPUTER WINNING GO

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TABLE OF CONTENTS

01

Introduction to The Go Game

History and Basic rules

02

UNINFORMED SEARCH

SIZE,
VERY SPECIFIC

03

MCT APPROACH

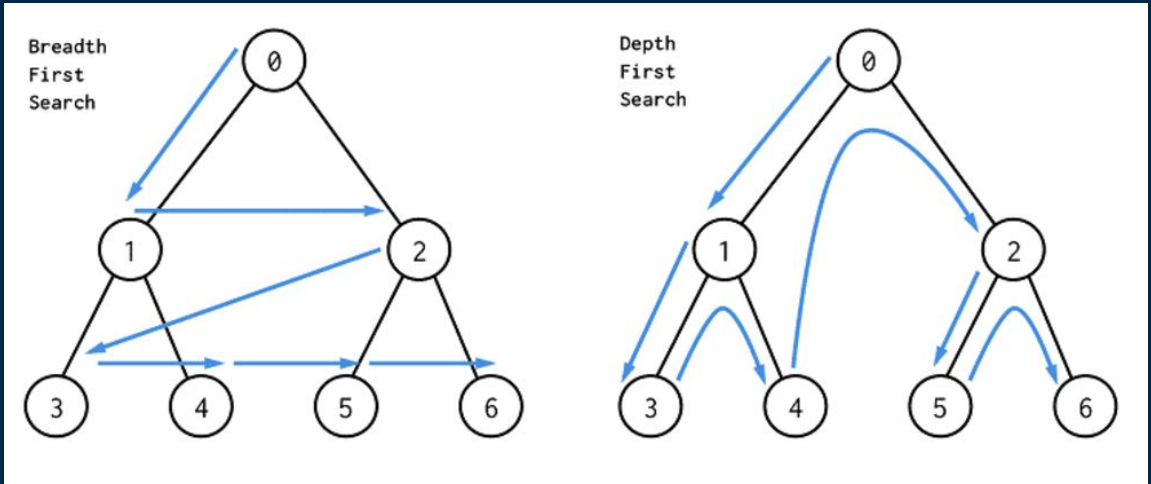
Selection
EXPANSION
STIMULATION
BACKUP

GAME OF GO

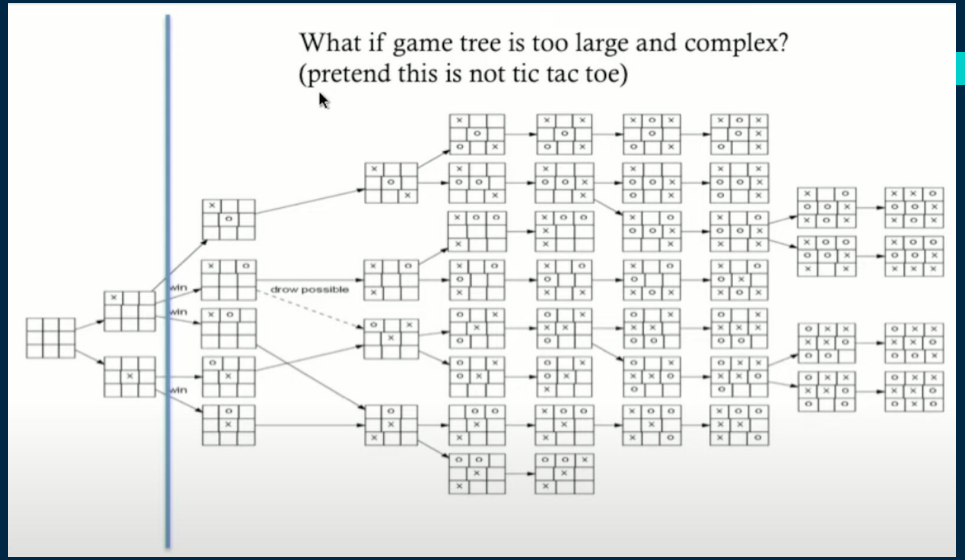
- 19*19 SIZE
- 2 PLAYER
- STARTS EMPTY BOARD
- GAIN MOST TERRITORY
- SUCCESSIVE PASS (LOSS)

UNINFORMED SEARCH

- GENERIC SEARCH ALGORITHM
- NO EXTRA INFORMATION
- TREE(BRANCH B,FACTOR D)= B^D



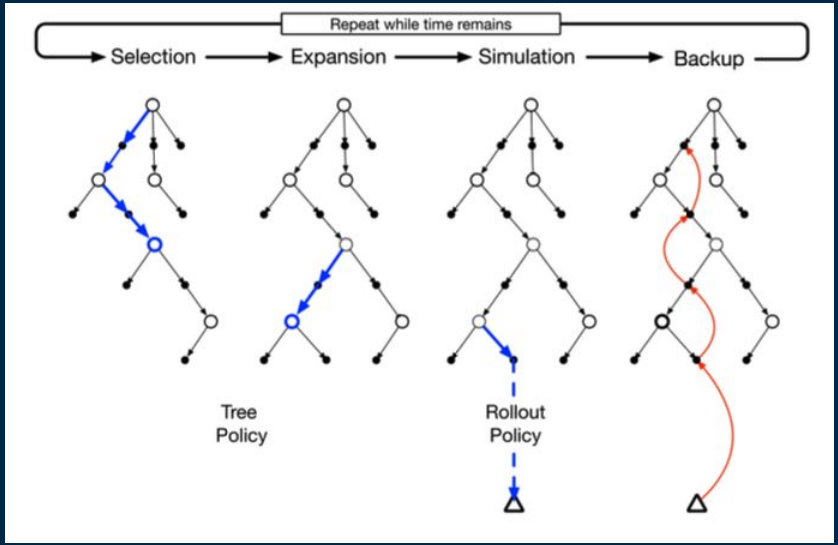
CONSEQUENCE



- At step 1: 250
- At step 2: 250^2 = 62500
- At step 3: 250^3 = 15,625,000
- At step 4: 250^4 = 3,906,250,000
- At step 5: 250^5 = 976,562,500,000
- ...
- At step 10: 250^{10} = 953,674,316,406,250,000,000,000
- NOTE:DUE TO LARGE SIZE IT STOPS AT BLUELINE(AI IS USELESS)

GOOGLE ALPHA GO AND MONTE CARLO TREE SEARCH

- SELECTION
- EXPANSION
- ROOLOUT/STIMULATION
- BACKUP

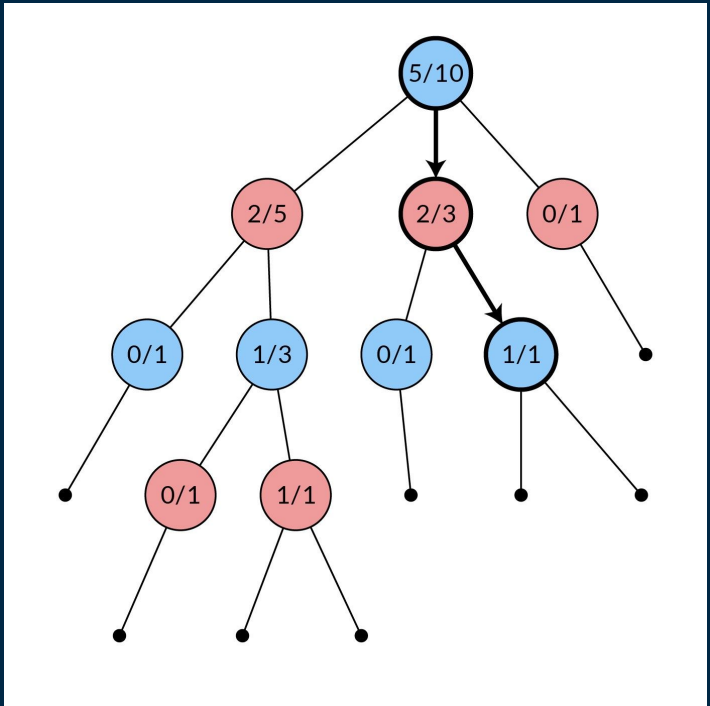


SELECTION

- START FROM ROOT NODE
- EXPLORE NEW PATHS(GAIN INFO)
- USE EXISTING INFO(EXPLOIT GOOD PATH)
- MOVE DOWN(LEGAL MOVES ONLY)
- STOP SELECTION(IF ILLEGAL MOVES)
- UPPER CONFIDENCE BOUND

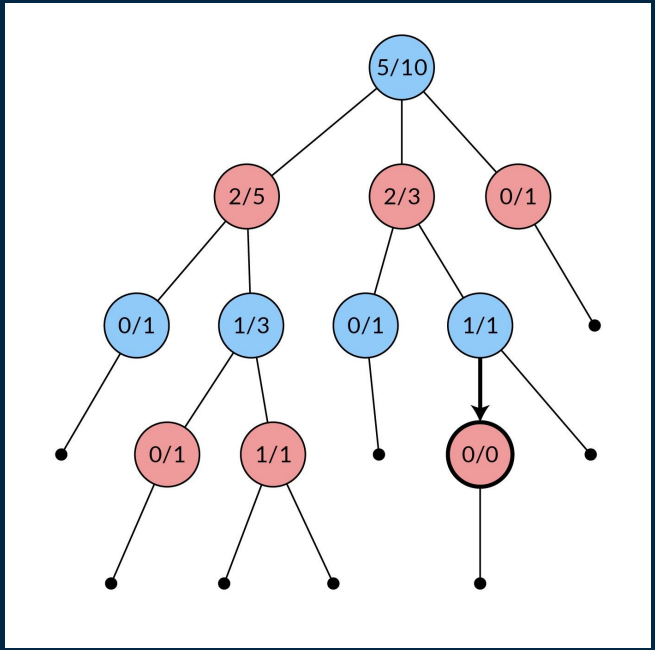
$$UCB(node_i) = \bar{x}_i + c\sqrt{\frac{\log N}{n_i}}$$

$\bar{x}_i$: mean node value; n_i: #visits of node i; N #visits parent;



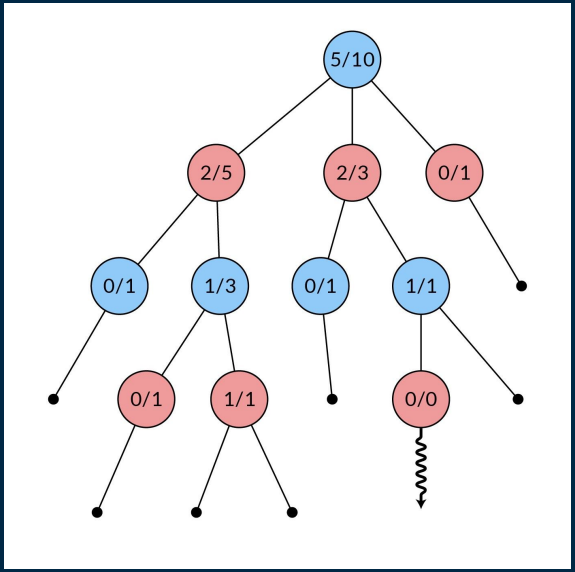
EXPANSION

- CHOOSE UNVISITED NODE
- SELECT UNEXPANDED NODE
- CREATE CHILD NOTE
- ADD CHILD TO LAST SELECTED NODE
- INITIALIZE STATISTICS INFO ($w_i = 0$, $s_i = 0$)



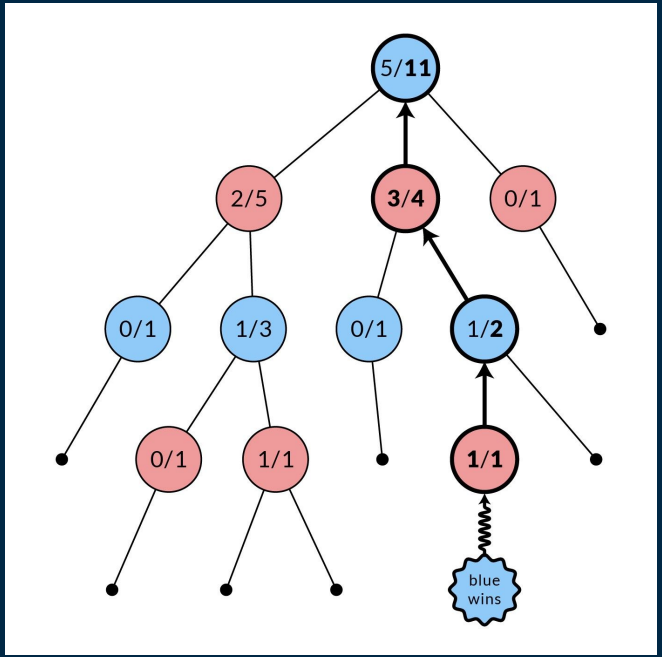
STIMULATION

- NEWLY CREATED NODE(CONT)
- MOVES SELECTED RANDOMLY
- UNTIL WINNER IS ASSIGNED
- NO NEW NODE
- IMPLEMENTING FOLLOWING RULES
    - FIND LEGAL MOVES
    - CHOOSE ONE(RANDOMLY)
    - ADVANCE GAME START
- GAME OVER



BACKPROPAGATION

- VISIT ALL STATICS NODES
- SIMULATION COUNT s_i INCREMENTED'
- UCB1 SELECTION FUNCTION FOR DECISION
- COMING FULL CIRCLE



THANK YOU

HAVE A GREAT DAY

CITATION

Chen, Keh-Hsun. "Maximizing the Chance of Winning in Searching Go Game Trees." Information sciences 175.4 (2005): 273–283. Print.

Gelly, Sylvain et al. "The Grand Challenge of Computer Go: Monte Carlo Tree Search and Extensions." Communications of the ACM 55.3 (2012): 106–113. Print.

Silver, David et al. "Mastering the Game of Go with Deep Neural Networks and Tree Search." Nature 529.7587 (2016): 484–489. Print.

Wang, Benjamin. "Monte Carlo Tree Search: An Introduction - towards Data Science." Towards Data Science. N.p., 10 Jan. 2021. Web. 27 April 2021.